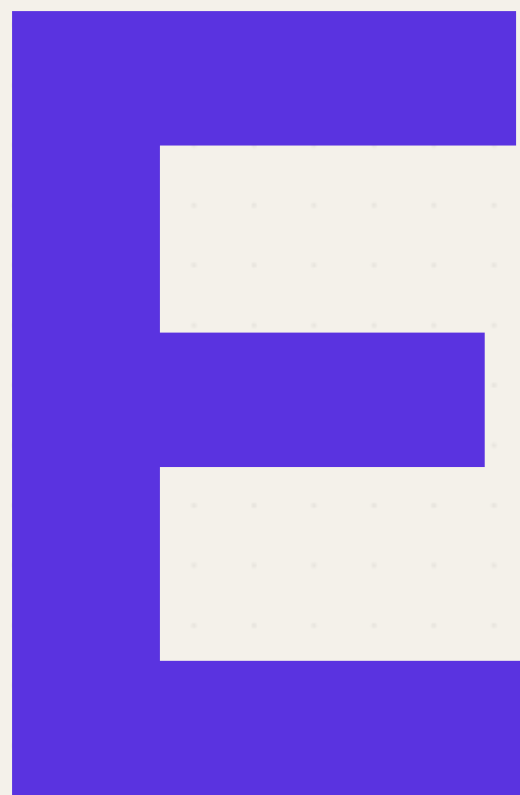




LV.0 → LV.1 · AI ENGINEER

The 6 E's that move you from **Level 0** → **1** AI engineer.

Ensemble, Entropy, Encoder-Decoder and more. **Plain-English** definitions, one swipe each, plus a researcher worth following.

EUCLIDEAN DISTANCE • FOUNDATIONS

Euclidean Distance

straight • line • distance

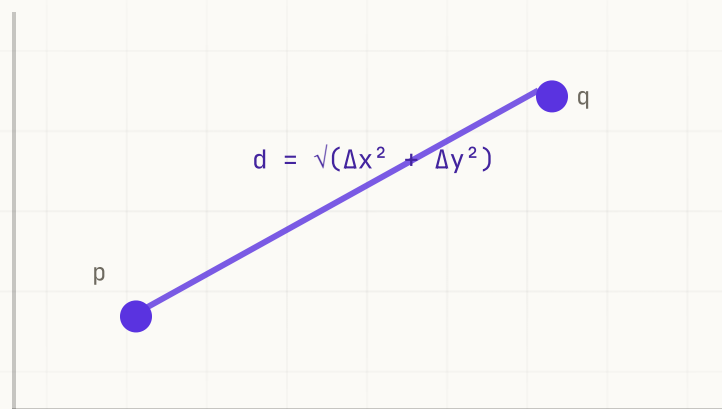
Euclidean distance is the **straight-line distance between two points**, computed with the Pythagorean theorem. In machine learning it measures how similar two data points or vectors are, powering algorithms such as K-Nearest Neighbors, K-Means clustering, and similarity search over embeddings.

USED IN

• KNN

• K-Means

• Similarity search



DISTANCE BETWEEN VECTORS

INFORMATION ENTROPY • FOUNDATIONS

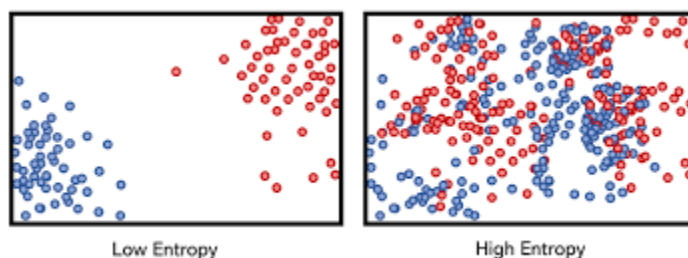
Entropy

measure • of • uncertainty

Entropy is a measure of uncertainty, randomness, or impurity in a dataset. In machine learning it quantifies **how mixed the class labels are** and is used in decision trees to choose the best feature for splitting data, with lower entropy meaning more homogeneous, informative partitions.

USED IN

- Decision trees
- Information gain
- Feature splitting



LOW VS HIGH ENTROPY

ENSEMBLE LEARNING • LEARNING

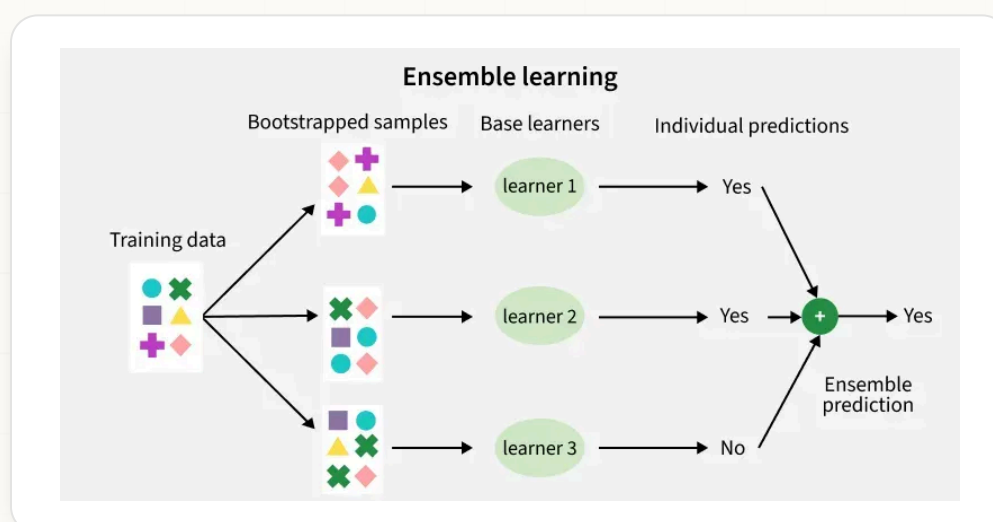
Ensemble

many • models • one vote

Ensemble learning is a technique that **combines the predictions of multiple models** to produce a more accurate, robust result than any single model. By leveraging diverse learners through bagging, boosting, and stacking, ensembles improve performance, reduce overfitting, and enhance generalization.

USED IN

- Bagging
- Boosting
- Random Forest



COMBINE MANY LEARNERS

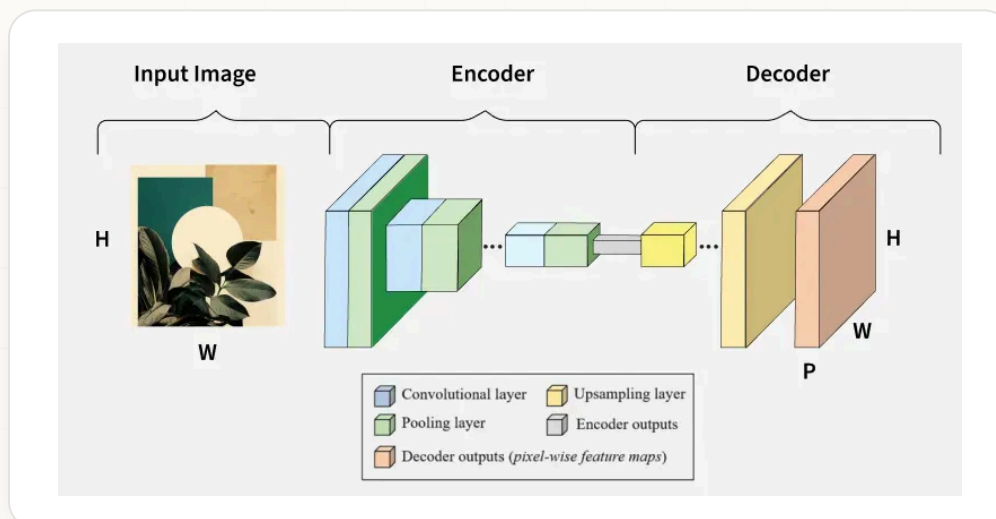
Encoder-Decoder

encode • compress • decode

An encoder-decoder architecture is a framework where **an encoder compresses input into a representation and a decoder turns it into output**. Widely used in machine translation, summarization, image captioning, and multimodal AI, it underpins models such as T5 and many sequence-to-sequence systems.

USED IN

- Translation
- Summarization
- Seq2seq



ENCODE → DECODE

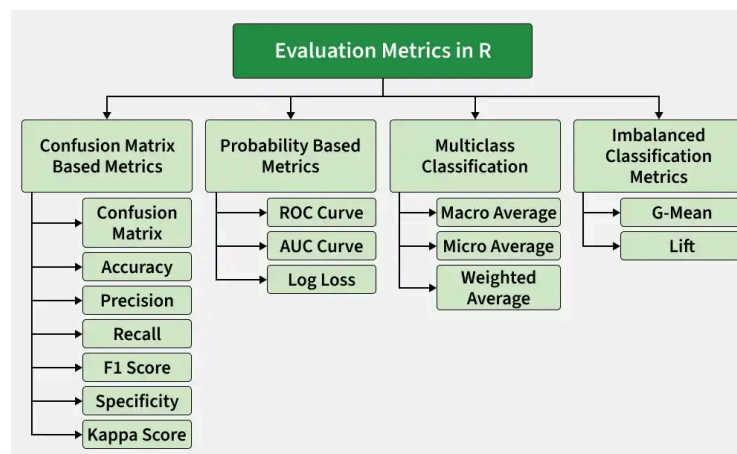
Evaluation Metrics

measure • compare • ship

Evaluation metrics are the **quantitative measures used to judge how well a model performs**. Classic ML uses accuracy, precision, recall, F1, and ROC-AUC, while GenAI adds faithfulness, answer relevance, groundedness, and LLM-as-judge scores, plus frameworks like RAGAS for evaluating retrieval and generation.

USED IN

- Accuracy & F1
- Faithfulness
- LLM-as-judge



FAMILIES OF METRICS

ENCRYPTION • SECURITY

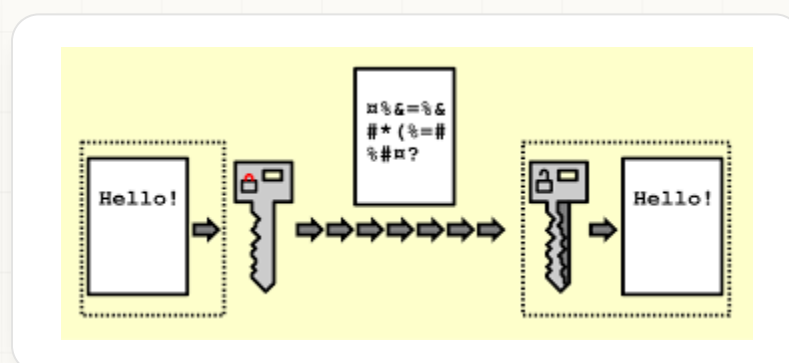
Encryption

encode • protect • decode

Encryption is the process of **converting readable data into an encoded format** using cryptographic algorithms and keys, so only authorized parties can access the original information. It is a fundamental pillar of cybersecurity, protecting sensitive data during storage, transmission, and communication across digital systems.

USED IN

- Cybersecurity
- Data privacy
- TLS



PLAINTEXT → CIPHERTEXT

↓ UP NEXT • PERSON TO FOLLOW

Spotlight next.

The **high-signal, low-noise** researcher worth a follow to keep your E-stack current.

E

HIGH-SIGNAL, LOW-NOISE

Edward Hu

AI researcher • LoRA, muP



Edward Hu is an AI researcher best known for **inventing LoRA, the low-rank adaptation method** that made fine-tuning large language models cheap and practical. His work on parameter-efficient training and muP scaling shapes how modern models are adapted and tuned across the industry.

[in /edwardjhu](#)

[X @edwardjhu](#)

WHAT THEY POST

Research on efficient training and scaling.

FIND THEM

LinkedIn and X [@edwardjhu](#).

WHY FOLLOW

The mind behind LoRA fine-tuning.

BEST FOR

Parameter-efficient fine-tuning ideas.

See you for F.

Save it, share it, and follow along. The 6 E's, plus a researcher worth following.

01 **Euclidean Distance** foundations

02 **Entropy** foundations

03 **Ensemble** learning

04 **Encoder-Decoder** architecture

05 **Evaluation Metrics** evaluation

06 **Encryption** security

+ **Edward Hu** people • lora



FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

in/prasanna-kulkarni-032650180

F Next